

Smartmeter / AMR LPO Documentation

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This documentation details the new flow for AMR / SmartMeter LPO data.

Kafka messages

In order to expose the data to the appropriate suppliers, the processed data is sent to each one via a kafka topic. First, some transformations are applied in order to format the data in order to comply with already existing schema from previous implementation. Each message from Kafka represents an Avro message. Besides the additional Kafka data provided, such as timestamp of message injection in Kafka, offset, and others, the data generated can be found in the following two keys:

1. KEY (contains the schema of the AVRO message)
2. VALUE (contains the values of the AVRO message)

The data is saved also in the 'PROCESSED' layer in the same S3 bucket before being sent on the kafka topic in order to be read by the supplier

Topic

cuifurnizor_lpo

Data format - avro schema

```
1 {
2   "type": "record",
3   "namespace": "rer_lpo",
4   "name": "rer_lpo",
5   "fields": [
6     {
7       "name": "readDate",
8       "type": "string",
9       "doc": "Timestamp of the reading. Europe/Bucharest TZ"
10    },
11    {
12      "name": "timestamp",
13      "type": "string",
14      "doc": "Execution timestamp. Europe/Bucharest TZ"
15    },
16    {
17      "name": "meterSerialNumber",
18      "type": "string",
19      "doc": "Meter Serial Number Identifier"
20    },
21    {
22      "name": "supplyPointId",
23      "type": "string",
24      "doc": "Code of Point Of Delivery"
25    },
26    {
27      "name": "energyType",
28      "type": "string",
29      "doc": "List of type of energy Measured"
30    },
31    {
32      "name": "sampleFrequency",
```

```

33         "type": "string",
34         "doc": "The sample granularity, programmed into the
meter."
35     },
36     {
37         "name": "complete",
38         "type": "string",
39         "doc": "Value that indicates that the Samples was
completed for a specific day"
40     },
41     {
42         "name": "sampleDate",
43         "type": "string",
44         "doc": "Date and time of the Sample. Europe/Bucharest TZ"
45     },
46     {
47         "name": "sampleValues",
48         "type": "string",
49         "doc": "Value of Sample."
50     },
51     {
52         "name": "cui_furnizor",
53         "type": "string",
54         "doc": "The Fiscal Identity Code (CUI) of the Supplier."
55     }
56 ]
57 }

```

AVRO kafka message schema

	Column	Type	Description	Example
1	readDate	string	Timestamp of the reading. Europe/Bucharest TZ	2025-10-28 00:00:00.000
2	timestamp	string	Execution timestamp. Europe/Bucharest TZ	2025-10-28 13:30:42.000
3	meterSerialNumber	string	Meter Serial Number Identifier	ABC123456789
4	supplyPointId	string	Code of Point Of Delivery	RO123456789
5	energyType	string	Type of Energy Measured. The possible values are: A1: Imported Active Energy (A+) A2: Exported Active Energy (A-) R1: Imported Inductive Reactive Energy (R+L) R2: Exported Inductive Reactive Energy (R-L) R3: Imported Capacitive Reactive Energy (R+C)	A1

[illegible]

				000\$0.160000\$0.120000\$ 0.160000\$0.120000\$0.160 000\$0.120000\$0.160000\$ 0.120000\$0.160000\$0.199 960\$0.359960\$0.279960\$ 0.199960\$0.199960\$0.1600 00\$0.199960\$0.160000\$0. 120000\$0.160000\$0.16000 0\$0.199960\$0.120000\$0.1 20000\$0.160000\$0.19996 0\$0.160000\$0.120000\$0.1 60000\$0.160000\$0.16000 0\$0.160000\$0.199960\$0.1 60000\$0.279960\$0.19996 0\$0.199960\$0.199960\$0.2 39960\$0.199960\$0.19996 0\$0.199960\$0.199960\$0.19 9960\$0.239960\$0.199960 \$0.199960\$0.199960"
1 0	cui_furnizor	string	The Fiscal Identity Code (CUI) of the Supplier.	RO11111111

What has changed since previous implementation

Effective changes when compared to previous implementation:

- removed dataValidation column since the information is not available in source
- removed rv values and adapted the new sampleValues format since the information is not available in source
- removed meterLastSampleDate column since the information is not available in source
- the values in sampleValues are now numeric type and can contain up to 6 decimals
- timestamp format changes from 2025-10-17T00:00:00.000+03:00 to 2025-10-17 00:00:00 (local Romania time)
- readDate is now rounded at day level
- there will always be 96 intervals no matter DST (so no 92 or 100 intervals)
- the sample values are now in kWh for SmartMeters

Output example of avro file:

```
1 {'key': {'type': "record", "namespace": "rer_lpo", "name": "rer_lpo",
"fields": [{"name": "readDate", "type": ["string", "null"]}, {"name":
"timestamp", "type": ["string", "null"]}, {"name": "meterSerialNumber",
"type": ["string", "null"]}, {"name": "supplyPointId", "type":
["string", "null"]}, {"name": "energyType", "type": ["string",
"null"]}, {"name": "sampleFrequency", "type": ["string", "null"]},
{"name": "complete", "type": ["string", "null"]}, {"name":
"sampleDate", "type": ["string", "null"]}, {"name": "sampleValues",
"type": ["string", "null"]}, {"name": "cui_furnizor", "type":
["string", "null"]}]}}|30', 'cui_furnizor': 'R011111111', 'value':
b'\x00&2025-11-10 15:03:56\x00&2025-11-10
```

```
15:30:37\x00"R0123456789\x00\x1eABC123456789\x00\x04A1\x00\x0415\x00\x0
8true\x00&2025-08-20
00:00:00\x00\x00\r0$.359960$0.239960$0.279960$0.239960$0.199960$0.19996
0$0.239960$0.239960$0.239960$0.239960$0.199960$0.239960$0.239960$0.2799
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0000$0.199960$0.199960$0.239960$0.160000$0.239960$0.199960$0.199960$0.1
60000$0.199960$0.160000$0.239960$0.199960$0.160000$0.279960$0.160000$0.
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60$0.199960$0.239960$0.199960$0.199960$0.199960$0.199960$0.199960$0.239
960$0.199960$0.199960$0.199960\x00\x14R0111111111'}
```